



Chapter S3G: Data Protection, Ethics & Social Issues

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1. Privacy and integrity of data

Privacy and integrity of data are important to protect personal information, build trust, ensure compliance with regulations, preserve confidentiality, and mitigate risks. Protecting privacy and maintaining data integrity is crucial for individuals, organisations, and service providers to establish trust, protect sensitive information, and comply with legal frameworks. By implementing robust data protection measures, cybersecurity practices, and privacy controls, the risks associated with unauthorised access, misuse, or compromise of data can be minimised.

2. Methods to protect data

To protect data, here are some key approaches and best practices.

Access Control	Implement strict access control measures to ensure that only authorised individuals can access sensitive data . This involves using strong passwords, two-factor authentication, and role-based access controls.
Encryption	Encrypt data at rest and in transit to preserve confidentiality if the data is stolen or lost . Encryption converts data into an unreadable format, and only authorised parties with the encryption key can decrypt and access the data. Employ strong encryption algorithms and secure key management practices.
Regular Backups	Conduct regular backups of important data and ensure it can be restored in case of accidental deletion, data corruption or system failure . Store backups securely and use off-site storage to protect against physical disasters.
Physical Security	Maintain the physical security of data by preventing unauthorised access , theft, or damage to servers and devices. Measures include restricted access to server rooms, surveillance systems, and secure storage for backup media.
Firewalls and Intrusion Detection Systems	Deploy firewalls and intrusion detection systems (IDS) to prevent and detect unauthorised network access . Firewalls restrict incoming and outgoing traffic based on security rules, while IDS monitors network activity and alerts administrators to suspicious or unauthorised access attempts.
Secure Software Development	Follow secure coding practices when developing software or applications that handle sensitive data. This includes input validation, output encoding, and protection against common vulnerabilities like SQL injection.
Security Awareness Training	Educate students and staff about data security best practices to reduce human error . Teach them about the importance of strong passwords, avoiding phishing attempts, and being cautious with personal data.
Patch Management	Keep software and systems up to date with the latest security patches and updates. Regularly apply patches to address vulnerabilities and exploits that could be used to gain unauthorised access to data.
Data Minimisation	Only collect and retain the minimum amount of data necessary for the intended purpose. Avoid unnecessary data storage to reduce the risk of data exposure in case of a security breach.
Incident Response Plan	Implement an incident response plan to rapidly contain breaches, limit damage, and restore systems quickly through defined response and recovery procedures.



3. Backup and Archive

Backup and archiving are essential strategies for protecting data and ensuring business continuity.

A backup is a duplicate of active data that can be quickly restored in case of accidental deletion, corruption, or system failure. Regular backups safeguard against hardware failures, cyberattacks, or natural disasters, allowing organisations to resume operations with minimal disruption.

Archiving, on the other hand, involves storing data for long-term retention, often on cost-effective storage media such as optical disks or compressed formats. Archiving frees up space on primary systems and ensures access to historical data for auditing, legal compliance, or reference purposes.

Both backups and archives should ideally be stored in offsite locations or cloud platforms, providing protection against physical damage, theft, or localised incidents. Implementing redundancy through multiple backups or distributed archiving further enhances data resilience.

Backup and archiving strategies should align with organisational data retention policies, which define how long data should be kept, the frequency of backups or archiving, and the criteria for secure deletion.

4. Version control and naming convention

Version control (e.g., Git) is a system for managing changes to files, documents, or datasets over time. It is essential for collaboration, accountability, and maintaining the integrity of content.

A comprehensive history of changes is maintained, providing an audit trail that allows tracking and reviewing modifications. This promotes accountability and helps to understand how content has evolved.

Version control enables seamless collaboration, allowing multiple individuals to work concurrently, merge changes, and resolve conflicts. The ability to roll back to previous versions supports iterative development, allowing continuous refinement and improvement of files or projects without losing previous work.



Naming convention

Naming conventions are standardised rules for naming files, folders, variables, functions, classes, and other resources. Consistent and meaningful names improve readability, organisation, and maintainability, making it easier to understand the purpose of a resource at a glance.

In programming, for example, camelCase is often used for variables and methods (e.g., myVariable, calculateTotal), while class names follow PascalCase (e.g., MyClass). Beyond programming, naming conventions are applied to file management, project documentation, websites, and collaborative resources, helping teams work efficiently, navigate resources easily, and maintain clarity across systems.

5. Personal Data Protection Act

The collection, storage, and use of personal data raise ethical concerns related to privacy, consent, and data security. These concerns have led to the introduction of data protection laws such as the Personal Data Protection Act (PDPA), which regulates how organisations collect, use, and disclose personal data.

The PDPA aims to balance technological development with the protection of individuals' privacy rights, ensuring organisations are accountable for their data handling practices while allowing individuals control over their personal information.

Under the PDPA, organisations must appoint a Data Protection Officer (DPO) to oversee compliance. The DPO ensures PDPA requirements are met, promotes data protection awareness, manages data-related complaints, identifies risks, and liaises with the Personal Data Protection Commission (PDPC) when necessary.

Four key PDPA obligations

- **Consent:** Organisations must obtain clear and informed consent before collecting, using, or disclosing personal data.
- **Purpose limitation:** Personal data must be collected only for specific and reasonable purposes.
- **Protection:** Organisations must implement security measures to prevent unauthorised access, disclosure, or loss of data.
- **Retention limitation:** Personal data should not be kept longer than necessary.

Failure to comply with the PDPA may result in investigations, financial penalties, reputational damage, legal consequences, and business disruption.



6. Code of ethics (conduct) of a Computing professional

Ethics in computing refers to professional rules of conduct guiding moral behaviour.

Professional bodies such as the **Association for Computing Machinery (ACM)** and the **Institute of Electrical and Electronics Engineers (IEEE)**, though based in the USA, have global influence. Their codes of ethics provide guidance and standards for professional conduct in the computing and technology fields, promoting ethical behavior and responsible practices worldwide. Similarly, the **Singapore Computer Society (SCS)** provides a Code of Ethics and Conduct for IT professionals in Singapore.

Across these organisations, there are shared areas of conduct that computing professionals are expected to follow. The codes are designed to maintain high ethical standards, promote trust and integrity in the profession, and ensure that public interest is central to professional decisions. These codes are not prescriptive “look-up tables” but offer guidance on proper professional behaviour.

No.	Areas of conduct	Examples
1.	Integrity and Honesty	Act ethically and truthfully; avoid conflicts of interest; do not overstate competence.
2.	Responsible Use of Technology	Use technology ethically; avoid malicious software; respect digital rights and AI ethics.
3.	Confidentiality and Privacy	Protect client, user, and organisational data; prevent unauthorised access or disclosure.
4.	Competence	Maintain and update professional knowledge; ensure work is of high quality.
5.	Professional Conduct	Treat colleagues, clients, and users with respect and fairness; promote diversity and inclusion.
6.	Respect for Intellectual Property	Uphold copyrights, patents, and licensing agreements; avoid unauthorised use.
7.	Public Interest and Social Responsibility	Consider societal impact; contribute positively to the community.
8.	Continuing Professional Development	Keep up with technology and ethical practices; support lifelong learning in the computing community.



7. Social issues

Social impact refers to how technology affects individuals, communities, and society. It influences lifestyle, communication, entertainment, and health.

Positive Social Impacts

- **Communication & Connectivity:** Enables staying in touch across distances; fosters collaboration and virtual communities.
- **Entertainment & Content Access:** Digital platforms provide personalised, on-demand access to movies, music, and TV globally.
- **Health & Well-being:** Apps and wearable tech can support health monitoring and personal well-being.
- **Convenience & Safety:** Mobile apps can locate lost devices or help in emergency situations.

Negative Social Impacts

- **Privacy Concerns:** Tracking, personal data collection, and targeted advertising without full consent.
- **Cyber Risks:** Cyberbullying, harassment, hate speech, and emotional harm via anonymous online communication.
- **Misinformation:** Rapid spread of false information on social media; erosion of trust and social division.
- **Addiction & Mental Health:** Excessive use can lead to social isolation, comparison stress, and anxiety.

Digital Divide

- **Access Inequalities:** Socioeconomic and educational factors limit access to technology.
- **Bridging Gaps:** Promoting digital inclusion via affordable access, training, and support for marginalised communities.

Legal Issues

- **Intellectual Property:** Respect for copyrights, patents, and digital content ownership.
- **Cybersecurity & Data Privacy:** Laws and regulations to protect information and digital assets.



8. Economic and workplace issues

Technology has significantly transformed workplaces and the economy, offering efficiency, flexibility, and innovation, but also creating challenges like job displacement.

Positive Impacts

- **Efficiency & Automation:** Streamlines processes and reduces manual tasks.
- **Collaboration & Flexibility:** Tools like video conferencing, instant messaging, and project management software enable remote teamwork and flexible work arrangements.
- **Data-Driven Decisions:** Advanced analytics, AI, and machine learning allow organisations to analyse data, predict trends, and optimise operations.
- **New Business Models & Jobs:** Technology enables new industries and roles.
 - Example: Food delivery platforms create jobs for riders/drivers while changing consumer behaviour.

Negative Impacts

- **Job Displacement:** Automation, AI, and robotics can replace traditional roles.
- **Skill Shifts:** Workers need to adapt, upskill, and embrace lifelong learning to remain employable.

9. Tutorial

1. A college will need to back up and archive its students' records.
 - (a) Define backup and archive. [2]
 - (b) Describe one suitable backup procedure and one archiving procedure for student records. [2]
 - (c) Explain two consequences if backups are not maintained. [2]
 - (d) Explain two consequences if archives are not performed. [2]
2. [9569/2025/1/5e]
The software team that develops the application also writes technical documentation. They use version control with naming conventions for the technical documentation.
 - (a) Give two benefits of using version control. [2]
 - (b) Explain the need for naming conventions. [2]
3. An employee leaves a company and brings source code they wrote to a new employer.
 - (a) State one ethical issue involved. [1]
 - (b) Explain why this action may be unethical even if the employee wrote the code. [2]
 - (c) State one possible consequence for the employee. [1]
4. A company accidentally emails customers' NRIC numbers to the wrong recipient.
 - (a) Identify two PDPA obligations breached. [2]
 - (b) State one action the DPO should take. [1]
 - (c) Explain one consequence for the organisation. [2]



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5. A software fault in a self-driving vehicle could cause accidents. Discuss the ethical responsibilities of the developer. [4]
6. Social media has improved communication more than it has harmed society. Discuss this statement. [4]
7. Explain one economic benefit and one economic drawback of automation in the workplace. [4]

8. [9569/SP/1/7]

Computing professionals are usually required to conduct themselves ethically. A company has asked you to write a code of conduct for the computing professionals employed by that company.

The areas of conduct that should be included in the company's computing professionals' code of conduct include:

- Integrity
- Responsibility
- Competence
- Professionalism

(a) Give one example of a rule that you would expect to be included in each area of conduct. Each rule given must be different. [4]

(b) For two of your rules, give an example of the unethical behaviour it is designed to prevent. [2]

9. [9569/2023/1/8]

A technician is testing the control system software for a nuclear power station before implementation. They find a fault that, if not corrected, could have catastrophic consequences. The technician reports the error to their manager, who is a computer professional. The manager's response is:

'The delivery of the system to the client on time is more important. The error can be fixed later, and a patch issued after the control system is installed and operational.'

Discuss whether the manager is abiding by the code of ethics of a computer professional. [5]

10. [9569/2025/1/7]

Autonomous taxis are unmanned vehicles that operate using Artificial Intelligence (AI). These taxis make decisions independently. For example, they navigate the quickest and safest route to pick up and drop off passengers at their destination.

Discuss the ethical issues that may be raised by the introduction of autonomous taxis. [6]



10. References

Personal Data Protection Act (PDPA)	https://www.pdpc.gov.sg/overview-of-pdpa/the-legislation/personal-data-protection-act
SingHealth PDPA	https://www.singhealth.com.sg/about-singhealth/corporate-governance/personal-data-protection-act
SCS Code of Conduct	https://www.scs.org.sg/membership/code-of-conduct
ACM Code of Ethics and Professional Conduct	https://www.acm.org/code-of-ethics
IPOS Copyright	https://www.ipos.gov.sg/about-ip/copyright/
Computer Misuse Act	https://sso.agc.gov.sg/Act/CMA1993
Cybersecurity Act	https://sso.agc.gov.sg/Acts-Supp/9-2018/